WORKED SOLUTION

STEP II 1993, Question 4

Pure • Skew Lines and Closest Approach

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s2-q4

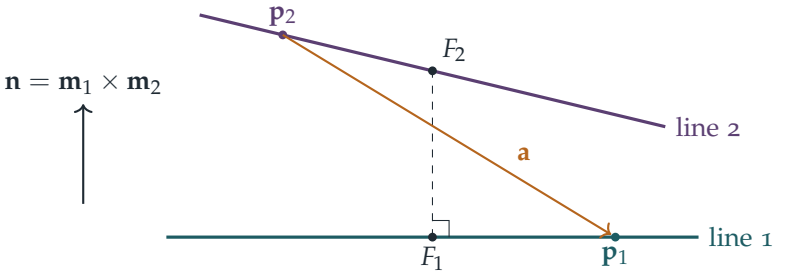
The key idea

Any vector joining a point of one line to a point of the other has the same component along the common normal $\mathbf{m}_1 \times \mathbf{m}_2$, because moving along either line adds only multiples of $\mathbf{m}_1$ and $\mathbf{m}_2$, which are perpendicular to that normal. That fixed component is the shortest distance. For the aircraft, the trick is to work in the frame of A_2: the vector from A_2 to A_1 moves along a *single* straight line, so the closest approach is the distance from a point to a line.

Solution

The shortest distance between the lines

Write $\mathbf{a} = \mathbf{p}_1 - \mathbf{p}_2$ and $\mathbf{n} = \mathbf{m}_1 \times \mathbf{m}_2$, which is non-zero because the lines are not parallel, and is perpendicular to both $\mathbf{m}_1$ and $\mathbf{m}_2$.



The sketch shows the two lines (skew, so they do not meet), the vector $\mathbf{a}$ from $\mathbf{p}_2$ to $\mathbf{p}_1$, and the common perpendicular F_1F_2, which is parallel to $\mathbf{n}$.

A general vector from a point of line 2 to a point of line 1 is

$$\mathbf{d} = \mathbf{a} + t_1\mathbf{m}_1 - t_2\mathbf{m}_2.$$

Since $\mathbf{m}_1 \cdot \mathbf{n} = \mathbf{m}_2 \cdot \mathbf{n} = 0$, its component along the unit normal $\hat{\mathbf{n}} = \mathbf{n}/|\mathbf{n}|$ is

$$\mathbf{d} \cdot \hat{\mathbf{n}} = \mathbf{a} \cdot \hat{\mathbf{n}},$$

the same for every choice of t_1, t_2. A vector is at least as long as any of its components, so $|\mathbf{d}| \geq |\mathbf{a} \cdot \hat{\mathbf{n}}|$, with equality exactly when $\mathbf{d}$ is parallel to $\mathbf{n}$. Equality does happen: $\mathbf{m}_1, \mathbf{m}_2, \mathbf{n}$ are not coplanar, so they form a basis and we can write $\mathbf{a} = \alpha\mathbf{m}_1 + \beta\mathbf{m}_2 + \gamma\mathbf{n}$; choosing $t_1 = -\alpha$, $t_2 = \beta$ leaves $\mathbf{d} = \gamma\mathbf{n}$. That choice gives the feet F_1, F_2 in the sketch. In the picture, the shortest distance is the "height" of $\mathbf{a}$ in the direction of $\mathbf{n}$. Hence the shortest distance is

$$|\mathbf{a} \cdot \hat{\mathbf{n}}| = \frac{|(\mathbf{p}_1 - \mathbf{p}_2) \cdot (\mathbf{m}_1 \times \mathbf{m}_2)|}{|\mathbf{m}_1 \times \mathbf{m}_2|}.$$

Part (i)

Here $\mathbf{a} = (1, 1, 1)$. The formula is unchanged if we replace $\mathbf{m}_1$ and $\mathbf{m}_2$ by any non-zero multiples, since the factors cancel top and bottom, so we may use $(4, 3, 0)$ and $(0, -3, 1)$:

$$(4, 3, 0) \times (0, -3, 1) = (3, -4, -12), \qquad |(3, -4, -12)| = \sqrt{9 + 16 + 144} = 13.$$

Then $(1, 1, 1) \cdot (3, -4, -12) = -13$, so the shortest distance is $13/13 = 1$.

Part (ii)

At time t the aircraft are at $\mathbf{p}_1 + v_1 t\, \mathbf{m}_1$ and $\mathbf{p}_2 + v_2 t\, \mathbf{m}_2$, so the vector from A_2 to A_1 is

$$\mathbf{r}(t) = \mathbf{a} + t\mathbf{w}, \qquad \mathbf{w} = v_1\mathbf{m}_1 - v_2\mathbf{m}_2.$$

Since $\mathbf{m}_1, \mathbf{m}_2$ are not parallel, $\mathbf{w} \neq \mathbf{0}$ (unless both speeds are zero). We take the flight to include all times, before and after $t = 0$. Then

$$|\mathbf{r}(t)|^2 = |\mathbf{a}|^2 + 2t\, \mathbf{a} \cdot \mathbf{w} + t^2 |\mathbf{w}|^2,$$

a quadratic in t with positive leading coefficient. Completing the square,

$$|\mathbf{r}(t)|^2 = |\mathbf{w}|^2 \Big(t + \frac{\mathbf{a} \cdot \mathbf{w}}{|\mathbf{w}|^2} \Big)^2 + |\mathbf{a}|^2 - \frac{(\mathbf{a} \cdot \mathbf{w})^2}{|\mathbf{w}|^2},$$

so the minimum, at $t = -\mathbf{a} \cdot \mathbf{w}/|\mathbf{w}|^2$, is

$$d^2 = \frac{|\mathbf{p}_1 - \mathbf{p}_2|^2\, |v_1\mathbf{m}_1 - v_2\mathbf{m}_2|^2 - [(\mathbf{p}_1 - \mathbf{p}_2) \cdot (v_1\mathbf{m}_1 - v_2\mathbf{m}_2)]^2}{|v_1\mathbf{m}_1 - v_2\mathbf{m}_2|^2},$$

as required. Geometrically, d is the distance from the origin to the line $\mathbf{r} = \mathbf{a} + t\mathbf{w}$, which is $|\mathbf{a}| \sin\theta$, where θ is the angle between $\mathbf{a}$ and $\mathbf{w}$.

Part (iii)

First a bound that needs no algebra: at every moment A_1 is on line 1 and A_2 on line 2, so $d \geq 1$ by part (i). Equality would need both aircraft to be at the feet F_1, F_2 *at the same time*.

Ask yourself Can the pilot of A_2 actually arrange that?

Find the feet. Requiring $\mathbf{a} + t_1\mathbf{m}_1 - t_2\mathbf{m}_2$ to be perpendicular to $\mathbf{m}_1$ and $\mathbf{m}_2$, with $\mathbf{a}\cdot\mathbf{m}_1 = \frac{7}{5}$, $\mathbf{a}\cdot\mathbf{m}_2 = -\frac{2}{\sqrt{10}}$ and $\mathbf{m}_1\cdot\mathbf{m}_2 = -\frac{9}{5\sqrt{10}}$, and solving the two linear equations gives

$$t_1 = -\tfrac{20}{13}, \qquad t_2 = \tfrac{\sqrt{10}}{13}.$$

A_1 is at F_1 at time $t_1/v_1 < 0$, but A_2 is at F_2 at time $t_2/v_2 > 0$ when $v_2 > 0$. These can never agree: F_1 lies behind $\mathbf{p}_1$ while F_2 lies ahead of $\mathbf{p}_2$. So with A_2 flying in the direction $\mathbf{m}_2$, the value $d = 1$ can't be reached. To see what can be reached, we use part (ii).

Put $v_2 = \sqrt{10}\,\lambda v_1$, so λ is a scaled speed ratio. Then

$$\mathbf{w} = v_1(\tfrac{1}{5}(4,3,0) - \lambda(0,-3,1)) = \tfrac{v_1}{5}(4,\ 3+15\lambda,\ -5\lambda),$$

and the factor $\frac{v_1}{5}$ cancels from the formula for d^2. With $|\mathbf{a}|^2 = 3$,

$$\mathbf{a}\cdot(4,3+15\lambda,-5\lambda) = 7+10\lambda, \qquad |(4,3+15\lambda,-5\lambda)|^2 = 25+90\lambda+250\lambda^2,$$

so

$$d^2 = 3 - \frac{(7+10\lambda)^2}{25+90\lambda+250\lambda^2} = \frac{26+130\lambda+650\lambda^2}{25+90\lambda+250\lambda^2} = \frac{26}{5}\cdot\frac{1+5\lambda+25\lambda^2}{5+18\lambda+50\lambda^2}.$$

Differentiating the last fraction N/D, the numerator of the derivative simplifies:

$$\begin{aligned} N'D - ND' &= (5+50\lambda)(5+18\lambda+50\lambda^2) - (1+5\lambda+25\lambda^2)(18+100\lambda) \\ &= 7+150\lambda+200\lambda^2 = (1+20\lambda)(7+10\lambda). \end{aligned}$$

So, as a function of every real λ, d^2 has a maximum at $\lambda = -\frac{7}{10}$ (where $\mathbf{a}\perp\mathbf{w}$ and $d = \sqrt{3}$) and a minimum at $\lambda = -\frac{1}{20}$, where

$$d^2 = \frac{26}{5}\cdot\frac{1-\frac{1}{4}+\frac{1}{16}}{5-\frac{9}{10}+\frac{1}{8}} = \frac{26}{5}\cdot\frac{13/16}{169/40} = 1.$$

This is the bound $d = 1$, attained when $v_2 = -\frac{v_1}{2\sqrt{10}}$, which agrees with t_2/t_1 from the feet.

So the aircraft can come within $d = 1$, the shortest distance between the lines, and no closer. To achieve it, the pilot of A_2 flies along line 2 in the direction $-\mathbf{m}_2$ at speed $\frac{v_1}{2\sqrt{10}}$: in the question's notation, $v_2 = -\frac{v_1}{2\sqrt{10}}$, a velocity along $\mathbf{m}_2$ rather than a speed. Then both aircraft reach the feet of the common perpendicular at the same moment. This reproduces the answer to part (i), which is presumably what the examiners intended. Notice that the answer depends only on the ratio v_2/v_1.

A remark on positive speeds. If v_2 must be a genuine speed, so $v_2 > 0$ and $\lambda > 0$, then $(1+20\lambda)(7+10\lambda) > 0$ and d^2 is strictly increasing in v_2:

$$d^2 \to \frac{26}{25} \quad \text{as } v_2 \to 0^+, \qquad d^2 \to \frac{13}{5} \quad \text{as } v_2 \to \infty.$$

In that reading there is no best speed: the pilot does better the slower they fly, and the closest approach can be made as near as we like to $\frac{\sqrt{26}}{5} \approx 1.020$, the distance from $\mathbf{p}_2$ to line 1, but never reaches it.

Checks

- The common perpendicular in (i): with $t_1 = -\frac{20}{13}$, $t_2 = \frac{\sqrt{10}}{13}$ the joining vector is $\frac{1}{13}(-3, 4, 12)$, which is parallel to $(3, -4, -12)$ and has length exactly 1.
- A computer minimised $|\mathbf{r}(t)|$ by direct search for random speeds and matched the formula in (ii) to 10^{-6}. It also scanned v_2/v_1 over $[-2, 2]$: the minimum is $d^2 = 1.0000$ at $v_2/v_1 \approx -0.1581 = -1/(2\sqrt{10})$, and d^2 rises from 1.04 for $v_2 > 0$.
- Limiting cases: $v_2 = 0$ makes A_2 a fixed point at $\mathbf{p}_2$, and $3 - (\frac{7}{5})^2 = \frac{26}{25}$ is indeed the squared distance from $\mathbf{p}_2$ to line 1.

Common slips

- Answering (iii) with "1, from part (i)" without checking that both aircraft can reach the feet of the common perpendicular at the same moment.
- Forgetting that the formula in (ii) is unchanged by scaling $\mathbf{w}$, and so carrying v_1 through pages of algebra.
- Dropping the modulus in the line-distance formula and getting -1.

Beyond STEP

Part (ii) is a small example of *orthogonal projection*: d is the length of the component of $\mathbf{a}$ perpendicular to $\mathbf{w}$, and $|\mathbf{a}|^2|\mathbf{w}|^2 - (\mathbf{a} \cdot \mathbf{w})^2 = |\mathbf{a} \times \mathbf{w}|^2$ is the Gram determinant, the squared area of the parallelogram on $\mathbf{a}$ and $\mathbf{w}$. Least-squares fitting in a first linear algebra course rests on the same projection idea. Strang's *Introduction to Linear Algebra* develops it beautifully.

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